

Sinusoids

- Sinusoids are periodic functions that can be represented as a sum of sine and cosine functions. The general form of a sinusoid is:

$$y(t) = A \sin(\omega t + \phi) + B \cos(\omega t + \phi) \quad (1)$$

where:

- A is the amplitude
 - ω is the angular frequency
 - ϕ is the phase shift
 - B is the vertical shift
- The angular frequency ω is related to the frequency f and the period T by:

$$\omega = 2\pi f = 2\pi \frac{1}{T} \quad (2)$$

- Hence, the sinusoid can also be expressed in terms of frequency:

$$y(t) = A \sin(2\pi f t + \phi) + B \cos(2\pi f t + \phi) \quad (3)$$

Discrete Time Sinusoids

- Discrete time sinusoids are sampled versions of continuous time sinusoids. The general form is:

$$x[n] = \sum_{k=0}^{K-1} a_k \sin(2\pi f_k t_n) + b_k \cos(2\pi f_k t_n) \quad (4)$$

where:

- n is the discrete time index
- k is the frequency index
- t_n is the discrete time variable $n\Delta t$
- f_k is the frequency of the k -th component $k\Delta f$
- Δt is the sampling period
- Δf is the frequency resolution
- K is the total number of frequencies

The coefficients a_k and b_k define the amplitude and phase of each frequency component.

- The discrete time sinusoid can also be expressed in different forms:

$$\begin{aligned} x[n] &= \sum_{k=0}^{K-1} a_k \sin(2\pi f_k t_n) + b_k \cos(2\pi f_k t_n) \\ &= \sum_{k=0}^{K-1} c_k \sin(2\pi f_k t_n + \phi_k) \\ &= \sum_{k=0}^{K-1} d_k \cos(2\pi f_k t_n + \theta_k) \end{aligned} \quad (5)$$

where:

- Second line is the same sinusoid as a summation of shifted sine functions
- Third line is the same sinusoid as a summation of shifted cosine functions

- c_k and d_k are the coefficients for the sine and cosine forms
- ϕ_k and θ_k are the phase shifts for the sine and cosine forms
- The relationship between the coefficients is given by:

$$\begin{aligned} c_k &= \sqrt{a_k^2 + b_k^2} \\ d_k &= \sqrt{a_k^2 + b_k^2} \\ \phi_k &= \tan^{-1} \left(\frac{b_k}{a_k} \right) \\ \theta_k &= \tan^{-1} \left(\frac{a_k}{b_k} \right) \end{aligned} \quad (6)$$

- If $a_k = b_k$, the sinusoid is a combination of sine and cosine functions with equal amplitude and phase shift. Phase shift is 45 degrees.
- If $a_k = -b_k$, the sinusoid is a combination of sine and cosine functions with equal amplitude and phase shift. Phase shift is 135 degrees.
- If $a_k = 0$, phase shift is 90 degrees
- If $b_k = 0$, phase shift is 0 degrees

Periodicity

- Fundamental period is the smallest period of a sinusoid. The fundamental period is the time it takes for the sinusoid to complete one full cycle.
- The fundamental frequency is the reciprocal of the fundamental period:

$$f_0 = \frac{1}{T_0} \quad (7)$$

- The fundamental frequency is the lowest frequency of a sinusoid. It is the frequency at which the sinusoid oscillates.
- Sinusoids are periodic functions, meaning they repeat their values at regular intervals.
- If we add sinusoids with different frequencies, the resulting function is not periodic unless the frequencies are rationally related.
- Suppose we have a sinusoid with the equation:

$$y(t) = a_1 \sin(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \quad (8)$$

- To determine if the sinusoid is periodic, we need to check if the ratio of the **frequencies** is rational:

$$\frac{\omega_1}{\omega_2} = \frac{p}{q} \quad (9)$$

where p and q are integers.

- To determine the **fundamental period** of the sinusoid, we need to find the least common multiple (LCM) of the periods of the individual sinusoids:

$$T = \text{lcm}(T_1, T_2) \quad (10)$$

where T_1 and T_2 are the periods of the individual sinusoids. Calculate the period of each sinusoid using:

$$T_i = \frac{2\pi}{\omega_i} \quad (11)$$

- Hint: Let ratio of periods = $\frac{T_1}{T_2} = \frac{q}{p}$, then $T = pT_1 = qT_2$

- Suppose we have a sinusoid with the equation:

$$y(t) = a_1 \sin(2\pi f_1 t + \phi_1) + a_2 \cos(2\pi f_2 t + \phi_2) \quad (12)$$

where f_1 and f_2 are integers

- To determine the **fundamental frequency** of the sinusoids, we need to find the greatest common divisor (GCD) of the frequencies of the individual sinusoids:

$$f = \text{gcd}(f_1, f_2) \quad (13)$$

- Then fundamental period is: $T = \frac{1}{f}$
- The above calculation can be done with sinusoids of more than 2 frequencies as well.

Fourier Series

Recap

- Orthogonal Basis Set** a set of functions that are orthogonal to each other. This means that the inner product of any two functions in the set is zero.
- Inner product of two functions $f(t)$ and $g(t)$ is defined as:

$$\langle f(t), g(t) \rangle = \int_{-\infty}^{\infty} f(t)g(t)dt \quad (14)$$

- Inner Product (continuous) = Dot Product (discrete)
- Sine and cosine functions are orthogonal to each other**, no matter their frequencies. Inner/Dot product = **zero**.

Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\begin{aligned} \int_0^T \sin(2\pi f_m t) \cos(2\pi f_n t) dt &= \frac{1}{2} \left[\frac{\cos(2\pi(f_m - f_n)t)}{-2\pi(f_m - f_n)} - \frac{\cos(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)} \right]_0^T \\ &= \begin{cases} 0 & \text{if } m \neq n \\ 0 & \text{if } m = n \end{cases} \end{aligned}$$

- Sine waves of different frequencies** are orthogonal to each other. Inner/Dot product = **zero**.

Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\begin{aligned} \int_0^T \sin(2\pi f_m t) \sin(2\pi f_n t) dt &= \frac{1}{2} \left[\frac{\sin(2\pi(f_m - f_n)t)}{2\pi(f_m - f_n)} - \frac{\sin(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)} \right]_0^T \\ &= \begin{cases} 0 & \text{if } m \neq n \\ \frac{T}{2} & \text{if } m = n \end{cases} \end{aligned}$$

- Similarly, **Cosine waves of different frequencies** are orthogonal to each other. Inner/Dot product = **zero**.

Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\begin{aligned} \int_0^T \cos(2\pi f_m t) \cos(2\pi f_n t) dt &= \frac{1}{2} \left[\frac{\sin(2\pi(f_m - f_n)t)}{2\pi(f_m - f_n)} + \frac{\sin(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)} \right]_0^T \\ &= \begin{cases} 0 & \text{if } m \neq n \\ \frac{T}{2} & \text{if } m = n \end{cases} \end{aligned}$$

- For sine and cosine with non-zero phase shifts:

$$\begin{aligned} \sin(2\pi f t + \phi) &= \sin(2\pi f t) \cos(\phi) + \cos(2\pi f t) \sin(\phi) \\ \cos(2\pi f t + \phi) &= \cos(2\pi f t) \cos(\phi) - \sin(2\pi f t) \sin(\phi) \end{aligned} \quad (15)$$

- In other words, with a phase shift, a sine or cosine function that is phase shifted can be expressed as a weighted sum of sine and cosine functions.
- So in general, the sum of sinusoidal signals, even with non-zero (but constant) phase, can be written as a linear combination of sine and cosine with zero phase.

$$\sum_{k=0}^{K-1} A_k \sin(2\pi f_k t + \phi_k) = \sum_{k=0}^{K-1} A_k [\sin(2\pi f_k t) \cos(\phi_k) + \cos(2\pi f_k t) \sin(\phi_k)]$$

$$\sum_{k=0}^{K-1} B_k \cos(2\pi f_k t + \phi_k) = \sum_{k=0}^{K-1} A_k [\cos(2\pi f_k t) \cos(\phi_k) - \sin(2\pi f_k t) \sin(\phi_k)]$$

Fourier Series

- Fourier series is a way to represent a periodic signal as a sum of sine and cosine functions. It is used to analyze periodic signals in the frequency domain.
- Sine and cosine functions make up the basis functions**. I.e for a sinusoidal signal with highest frequency f_m , we can represent it as a sum of M sine and cosine functions with frequencies $f_0, f_1, \dots, f_{M-1}$.

$$y(t) = \sum_{m=0}^{M-1} (a_m \sin(2\pi f_m t) + b_m \cos(2\pi f_m t)) \quad (16)$$

- How do we determine all of the a_m and b_m :

Recall that we can represent any periodic signal by a linear combination of shifted sines, i.e.

$$\begin{aligned} \text{any periodic signal} &= \sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m) \\ &= \sum_{m=0}^{M-1} A_m (\sin(2\pi f_m t) \cos(\phi_m) + \cos(2\pi f_m t) \sin(\phi_m)) \\ &= \sum_{m=0}^{M-1} [A_m \cos(\phi_m) \sin(2\pi f_m t) + A_m \sin(\phi_m) \cos(2\pi f_m t)] \\ &= \sum_{m=0}^{M-1} \underbrace{A_m \cos(\phi_m)}_{a_m} \sin(2\pi f_m t) + \sum_{m=0}^{M-1} \underbrace{A_m \sin(\phi_m)}_{b_m} \cos(2\pi f_m t) \end{aligned}$$

$$\begin{aligned} a_m &= A_m \cos(\phi_m) \\ b_m &= A_m \sin(\phi_m) \end{aligned} \quad (17)$$

- The Fourier series coefficients a_m and b_m are calculated using the following formulas:

$$\begin{aligned} a_m &= \frac{2}{T} \int_0^T y(t) \sin(2\pi f_m t) dt \\ b_m &= \frac{2}{T} \int_0^T y(t) \cos(2\pi f_m t) dt \end{aligned} \quad (18)$$

Recall the following integration results (Dot Product)

$$\begin{aligned} \int_0^T \left[\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m) \right] \sin(2\pi f_k t) dt &= \frac{T}{2} A_k \cos(\phi_k) \\ \int_0^T \left[\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m) \right] \cos(2\pi f_k t) dt &= \frac{T}{2} A_k \sin(\phi_k) \\ \int_0^T \left[\sum_{m=0}^{M-1} B_m \cos(2\pi f_m t + \phi_m) \right] \sin(2\pi f_k t) dt &= -\frac{T}{2} A_k \sin(\phi_k) \\ \int_0^T \left[\sum_{m=0}^{M-1} B_m \cos(2\pi f_m t + \phi_m) \right] \cos(2\pi f_k t) dt &= \frac{T}{2} A_k \cos(\phi_k) \end{aligned}$$

We can therefore use $\sin(2\pi f_k t)$ and $\cos(2\pi f_k t)$ to figure out what are all the a_m and b_m in the signal.

$$\int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \sin(2\pi f_k t) dt = \frac{T}{2} A_k \cos(\phi_k) = \frac{T}{2} a_k$$

$$\int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \cos(2\pi f_k t) dt = \frac{T}{2} A_k \sin(\phi_k) = \frac{T}{2} b_k$$

The ratio of b_k to a_k gives the phase shift of the sinusoid:

$$\frac{b_k}{a_k} = \frac{A_k \sin(\phi_k)}{A_k \cos(\phi_k)} = \tan(\phi_k)$$

$$\phi_k = \tan^{-1} \left(\frac{b_k}{a_k} \right)$$
(19)

We can use complex numbers to represent the Fourier series:

$$X(f_k) = \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \cos(2\pi f_k t) dt$$

$$-j \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \sin(2\pi f_k t) dt$$

$$= \frac{T}{2} A_k \sin(\phi_k) - j \frac{T}{2} A_k \cos(\phi_k)$$

$$= \frac{T}{2} (b_k - j a_k)$$

Magnitude = $|X_k| = \frac{T}{2} \sqrt{a_k^2 + b_k^2}$

Phase = $\tan^{-1} \left(\frac{b_k}{a_k} \right)$

We can also express the Fourier series in terms of **complex exponentials**, recall Eulers formula:
 $e^{j\theta} = \cos(\theta) + j \sin(\theta)$, then:

$$X(f_k) = \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] e^{-j2\pi f_k t} dt$$
(20)

Fourier series formula (important):

$$X(f_k) = \int_0^T x(t) e^{-j2\pi f_k t} dt$$
(21)

Fourier Transform

Fourier transform is a way to represent a non-periodic signal as a sum of sine and cosine functions. It is used to analyze non-periodic signals in the frequency domain.

Intuition: For non-periodic signals, we can think that its *period is infinite*. Then the whole signal can be imagined as periodic with infinite period.

Fourier series comparison:

$$X(f) = \int_0^T x(t) e^{-j2\pi f t} dt$$
(22)

Fourier transform is defined as:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$
(23)

The inverse Fourier transform is defined as:

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$$
(24)

The Fourier transform can be used to analyze the frequency content of a signal. It can be used to determine the amplitude and phase of each frequency component in the signal.

Fourier transform equation using radian frequency ω :

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$
(25)

The inverse Fourier transform using radian frequency ω :

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$
(26)

The Fourier transform can also be used to filter signals. By multiplying the Fourier transform of a signal by a filter function, we can remove or enhance certain frequency components in the signal.

Discrete Fourier Transform (DFT)

DFT Formula:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi k \frac{n}{N}}$$
(27)

Inverse DFT Formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi k \frac{n}{N}}$$
(28)

where:

- N is the number of samples
- k is the frequency index (discretised frequencies)
- n is the time index
- $x[n]$ is the input signal
- $X[k]$ is the DFT of the input signal

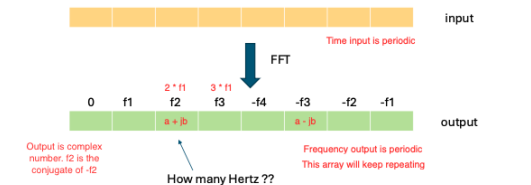
FFT (Fast Fourier Transform) is an efficient algorithm to compute the DFT. It reduces the computational complexity from $O(N^2)$ to $O(N \log N)$.

Frequency Bins

Frequency resolution: Smallest frequency difference that can be resolved by the DFT. It is determined by the sampling frequency and the number of samples.

The DFT maps the input signal to a set of frequency bins. Each bin corresponds to a specific frequency range.

When you input a vector of, say, 8 numbers into an FFT function, and you get an output vector of length 8, how do you interpret the output vector?



The frequency resolution of the DFT is given by:

$$\Delta f = \frac{f_s}{N}$$
(29)

where:

- f_s is the sampling frequency
- N is the number of samples

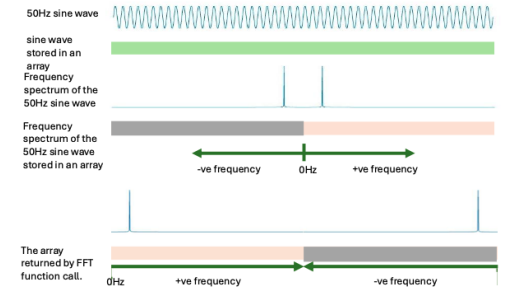
The frequency bins are given by:

$$f_k = k \Delta f$$
(30)

The DC component (k=0) corresponds to the average value of the signal. It is a constant in the sinusoid equation. The Nyquist frequency (k=N/2) corresponds to the highest frequency that can be represented by the DFT.

We can imagine the DC component as vertical offset of the signal from the x-axis.

Visualisation:



Derivation of the frequency bins:

$$H(f) = \int_{t=-\infty}^{\infty} h(t) e^{-j2\pi f t} dt$$

$$H(f) \approx \sum_{n=0}^{N-1} h[n] e^{-j2\pi f n \Delta t} \Delta t$$

$$H(f) \approx \Delta t \sum_{n=0}^{N-1} h[n] e^{-j2\pi f n \Delta t}$$

$$H(f) \approx \frac{1}{f_s} \sum_{n=0}^{N-1} h[n] e^{-j2\pi \frac{f}{f_s} n}$$

Let $f = k \Delta f$, i.e. we also discretize the frequency axis using a frequency step Δf

$$H[k] \approx \frac{1}{f_s} \sum_{n=0}^{N-1} h[n] e^{-j2\pi \frac{k \Delta f}{f_s} n}$$

Recall that $H[k] \approx \sum_{n=0}^{N-1} h[n] e^{-j2\pi \frac{k}{N} n}$

Ignoring the scale factor, we have

$$2\pi \frac{k \Delta f}{f_s} n = 2\pi \frac{k}{N} n$$

Therefore,

$$\frac{\Delta f}{f_s} = \frac{1}{N}$$

Note: Sometimes, we pad the input signal with zeros to increase the number of samples. Padding zeros to the input decreases the frequency bin size, but does not increase the frequency resolution.

Note: $N = 2^n$ for FFT, where n is an integer. This is because the FFT algorithm is based on the divide-and-conquer approach, which works best when the number of samples is a power of 2.

Properties of DFT

Intuition: When you discretise one domain, you also discretise the other domain. The *other domain becomes periodic*.

- Linearity:** $\mathcal{F}\{ax[n] + by[n]\} = aX[k] + bY[k]$
- Time Shifting:** $\mathcal{F}\{x[n - n_0]\} = X[k] e^{-j2\pi k \frac{n_0}{N}}$
- If you shift a signal $x[n]$ by m samples in time, the Fourier transform $X[k]$ remains the same in magnitude, but is multiplied by a complex exponential.
- This complex exponential introduces a linear phase shift in the frequency domain. This exponential has a magnitude of 1, so the magnitude spectrum stays unchanged.
- Frequency Shifting:** $\mathcal{F}\{x[n] e^{j2\pi m \frac{n}{N}}\} = X[k - m]$
- If you multiply $x[n]$ by a complex exponential in the time domain, it causes a circular shift in the frequency domain. Shifting the frequency response by m bins. Both magnitude and phase are affected.
- Time Reversal:** $\mathcal{F}\{x[-n]\} = X^*[k]$
- Conjugate Symmetry:** $X[k] = X^*[N - k]$
- Here $X^*[k]$ is the complex conjugate of $X[k]$. (i.e If $X[k] = a + jb$, then $X^*[k] = a - jb$)
- It is important to note that the $|X[k]| = |X^*[k]|$

Conversion **from continuous to discrete**:

$$\text{Let } t = n \cdot \Delta t = n \cdot \frac{T}{N} \quad \text{where } N \text{ is the total number of points}$$

$$-\infty < t < \infty \quad \rightarrow \quad -\infty < n < \infty$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \rightarrow \quad X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n \Delta t} \Delta t$$

$$= \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega \frac{nT}{N}}$$

Note: $\Delta t = \frac{T}{N}$

$$\text{Let } \omega = k \omega_0 = k 2\pi f_0 = k 2\pi \frac{1}{T}$$

$$X[k] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk2\pi \frac{1nT}{NT}}$$

$$X[k] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi k \frac{n}{N}}$$

DFT output is **periodic with period N** (repeats every N samples). $X[k] = X[k + N]$

$$X[k] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi k \frac{n}{N}}$$

$$X[k+N] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi(k+N)\frac{n}{N}}$$

$$= \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi k \frac{n}{N} - j2\pi n}$$

$$= \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi k \frac{n}{N}}$$

$$= X[k]$$

$\rightarrow X[k]$ is periodic with period N

• Similarly, the input signal is periodic with period N.
 $x[n] = x[n+N]$

$$x[n] = \frac{1}{N} \sum_{k=-\infty}^{\infty} X[k] e^{j2\pi k \frac{n}{N}} \quad (31)$$

• Summary:

continuous time, continuous frequency

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

CTFT
Continuous Time Fourier Transform

discrete time, continuous frequency (i.e. ω is continuous)

$$X(\omega) = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n \frac{T}{N}}$$

DTFT
Discrete Time Fourier Transform
Time axis is discretized but frequency axis is not

discrete time, discrete frequency (i.e. k is discrete)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi k \frac{n}{N}}$$

DFT (FFT)
Discrete Fourier Transform (DFT)
(Fast Fourier Transform is a fast implementation of DFT)

continuous time, continuous frequency

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Inverse CTFT

discrete time, continuous frequency (i.e. k is continuous)

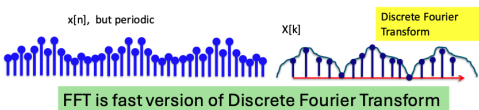
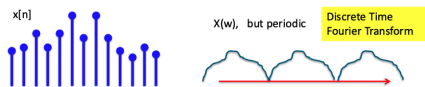
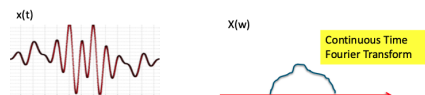
$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n \frac{T}{N}} d\omega$$

Inverse DTFT

discrete time, discrete frequency (i.e. k is discrete)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi k \frac{n}{N}}$$

Inverse DFT (IFFT)



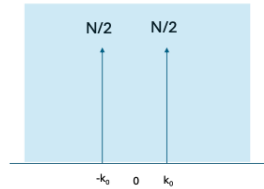
Useful Fourier Transform Results

- Fourier transform of an **impulse** in the time domain will result in a constant function in the frequency domain.
 $[1, 0, 0, 0] \rightarrow [1, 1, 1, 1]$
- Fourier transform of a **constant function** in the time domain will result in an impulse in the frequency domain.
 $[1, 1, 1, 1] \rightarrow [4, 0, 0, 0]$

Discrete Fourier Transform Pairs

$$\cos\left(\frac{2\pi}{N} k_0 n\right) \xleftrightarrow{\text{FFT}} \frac{N}{2} (\delta[k - k_0] + \delta[k + k_0])$$

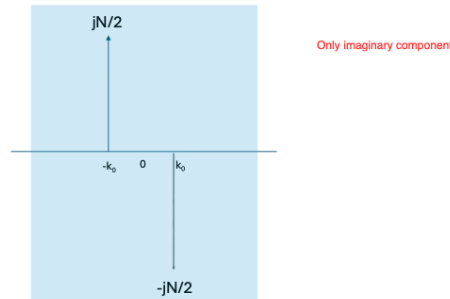
IFFT



Discrete Fourier Transform Pairs

$$\sin\left(\frac{2\pi}{N} k_0 n\right) \xleftrightarrow{\text{FFT}} \frac{-jN}{2} \delta[k - k_0] + \frac{jN}{2} \delta[k + k_0]$$

IFFT



Continuous Time Fourier Transform for an Impulse Train

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow P(\omega) = \sum_{k=-\infty}^{\infty} \omega_0 \delta(\omega - k\omega_0)$$

$$\text{where } \omega_0 = \frac{2\pi}{T_s}$$

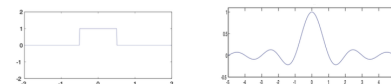


Continuous Time Fourier Transform for a Rectangular Function

$$\text{rect}(t) = \begin{cases} 1 & -0.5 \leq t \leq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

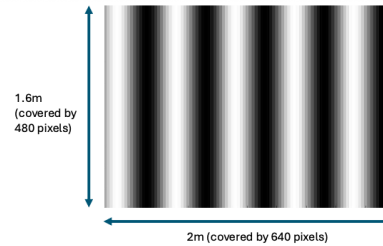
L'Hopital's Rule

$$x(t) = \text{rect}(t) \longleftrightarrow X(\omega) = \frac{2\sin(\frac{\omega}{2})}{\omega} = \text{sinc}(\frac{\omega}{2\pi})$$



2D Fourier Transform

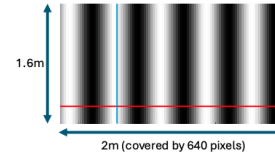
- The 2D Fourier transform is used to analyse 2D signals, such as images. Units: **cycles per meter**.
- Sampling frequency: f_s = Number of points/pixels, N per metre
- Frequency bin size: $\Delta f = \frac{f_s}{N}$



Suppose this scene measuring 1.6m by 2m is imaged by a camera of 480 pixels by 640 pixels. pixel $\rightarrow$ a point in the wave

In the horizontal direction, the sampling frequency f_s is 640/2 = 320 samples per metre

In the vertical direction, the sampling frequency f_s is 480/1.6 = 300 samples per metre



Sample 640 along red line and perform FFT.
Horizontal frequency bin size is given by

$$\text{horizontal } f_{res} = \frac{f_s}{640} = \frac{320}{640} = 0.5 \text{ Hz}$$

Sample 480 along orange line and perform FFT.
Vertical frequency bin size is given by

$$\text{vertical } f_{res} = \frac{f_s}{480} = \frac{300}{480} = 0.625 \text{ Hz}$$

What about frequencies along other lines and frequencies along directions not horizontal or vertical?
We will need 2D Fourier Transform (2D FFT).

- Intuition:** Applying 1D Fourier transform to each row of the image, then applying 1D Fourier transform to each column of the result.
- 2D Fourier transform (spatial domain to frequency domain):

$$G[k, l] = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} g[u, v] e^{-j2\pi(\frac{ku}{M} + \frac{lv}{N})} \quad (32)$$

- 2D inverse Fourier transform (frequency domain to spatial domain):

$$g[u, v] = \frac{1}{MN} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} G[k, l] \cdot e^{j2\pi(\frac{ku}{M} + \frac{lv}{N})} \quad (33)$$

where:

- $g[u, v]$ is the input image
- $G[k, l]$ is the 2D Fourier transform of the input image
- M and N are the dimensions of the image (horizontal and vertical respectively)
- k and l are the frequency indices
- u and v are the spatial indices
- For 2D DFT results with DC component at the top left corner, the top row represents horizontal frequencies, and the left column represents vertical frequencies.
- For 2D DFT results, the output is **periodic in both dimensions**. The output is a 2D array of complex numbers, where each element represents the amplitude and phase of a specific frequency component in the image.

- Frequency at peak** = (Coordinate at peak - Coordinate of DC component) $\times$ Frequency bin size

Convolution

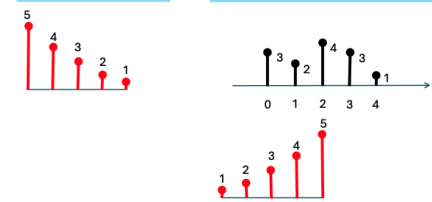
- Convolution is a mathematical operation that combines two functions to produce a third function.
- Convolution is used to filter signals and images. It is used to apply a filter to a signal or image to enhance or suppress certain features.

1D Convolution

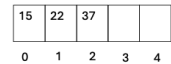
- Process of sliding a window (kernel) over a signal, and computing the weighted sum of the signal values within the window.
- The kernel defines the weights, and it is the **flipped** version of impulse response.

Impulse response of gong

Strength of hitting the gong at different time steps



What you will hear

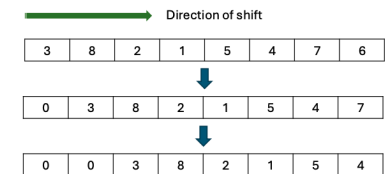


- Think of the hitting pattern as the signal, and the impulse response as the kernel.

Linear Convolution

Linear Convolution

- What we had done so far is linear convolution
- In linear convolution, the values in an array gets "shifted out"



- Linear convolution can be done in the frequency domain using the Fourier Transform.
- Linear shift invariance property:** The output of a linear time-invariant system is the convolution of the input signal with the impulse response of the system.
- The convolution theorem states that the **Fourier transform of the convolution of two signals is equal to the product of their Fourier transforms**:

$$\mathcal{F}\{x[n] * h[n]\} = X[k]H[k] \quad (34)$$

- Derivation of the convolution theorem (discrete version):

$$\begin{aligned}
 G[k] &= \mathcal{F}\{g[m]\} = \mathcal{F}\{x[m] * h[m]\} \\
 &= \sum_{m=0}^{M-1} \left(\sum_{\tau=0}^{M-1} x[\tau] h[m-\tau] \right) e^{-j2\pi k \frac{m}{M}} \\
 &= \sum_{\tau=0}^{M-1} x[\tau] \left(\sum_{m=0}^{M-1} h[m-\tau] e^{-j2\pi k \frac{m}{M}} \right) \\
 &= \sum_{\tau=0}^{M-1} x[\tau] H[k] e^{-j2\pi k \frac{\tau}{M}} \\
 &= H[k] \sum_{\tau=0}^{M-1} x[\tau] e^{-j2\pi k \frac{\tau}{M}} \\
 &= H[k] X[k]
 \end{aligned}$$

(35)

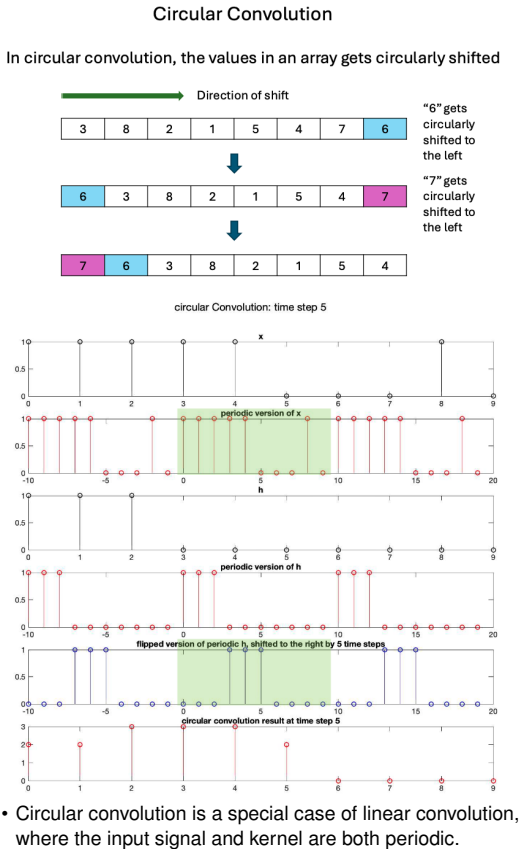
- This uses the circular shift property of DFT, where:

$$\mathcal{F}\{h[n-m]\} = H[k]e^{-j2\pi k \frac{m}{N}}$$

(36)

- Note: `ifft(fft(h(t)) .* fft(x(t)))`
- Length of the output signal** is $N + M - 1$, where N is the length of the input signal and M is the length of the kernel.

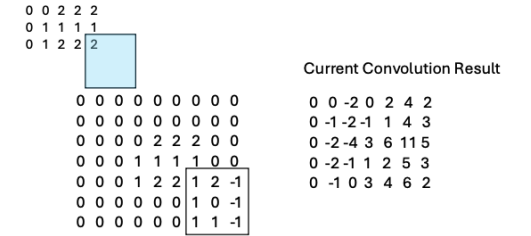
Circular Convolution



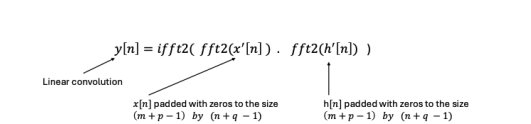
- To achieve circular convolution, we need to ensure that the input signal and kernel are both periodic with the same period. This is done by **zero-padding** the input signal and kernel to the same length.
- Normally circular convolution is "unwanted" because it can cause aliasing and distortion in the output signal.
- To prevent circular convolution, we can use zero-padding by adding zeros to the input signal and kernel before performing the convolution.
- We pad $x[n]$ with $M - 1$ zeros, and $h[n]$ with $N - 1$ zeros. This ensures that the output signal has a length of $N + M - 1$.

2D Convolution

- 2D convolution is used to filter images. It is similar to 1D convolution, but it uses a 2D kernel (filter) that slides over the image.
- The input image is treated as a 2D array with zeros padded around the edges.
- The kernel is flipped in both dimensions before convolution. (left-right and up-down)
- The kernel is applied to each pixel in the image, and the output pixel is computed as the weighted sum of the pixels in the kernel.
- Start from the top left corner of the image, and slide the kernel over the image.



- To implement linear convolution using FFT, we will need to handle the periodicity assumption of the DFT. This is again done by **zero-padding the input image and kernel**.
In general, if image $x[n]$ is of size m by n , and the kernel $h[n]$ is of size p by q , then both the image and kernel must be padded with zeros to the minimum size of $(m + p - 1)$ by $(n + q - 1)$ before performing the 2D FFT.



- $$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] \cos \left(\frac{\pi}{N} \left(n + \frac{1}{2} \right) k \right) \\ &= \sum_{n=0}^{N-1} x[n] \cos \left(\frac{\pi}{N} nk + \frac{\pi}{2N} k \right) \end{aligned} \quad (39)$$

-

- Quantization means dividing the DCT coefficients by a quantization matrix and rounding the result to the nearest integer. This reduces the number of bits needed to represent the DCT coefficients.
- Luminance channels are quantized with a different matrix than chrominance channels.
- Apply **zig-zag scanning** to the quantized DCT coefficients. This reorders the coefficients in a way that groups the low frequency components together. (So that RLE and Huffman coding can have a higher compression ratio).

0	1	5	6	14	15	27	28
2	4	7	13	16	26	29	42
3	8	12	17	25	30	41	43
9	11	18	24	31	40	44	53
10	19	23	32	39	45	52	54
20	22	33	38	46	51	55	60
21	34	37	47	50	56	59	61
35	36	48	49	57	58	62	63

- | Code | Frequency | Huffman Code | a: | 14 * 3 bits = 42 bits |
|-------|-----------|--------------|---------------------------------|-----------------------|
| a | 14 | 000 | b: | 1 * 6 bits = 6 bits |
| b | 1 | 111010 | c: | 2 * 5 bits = 10 bits |
| c | 2 | 11110 | d: | 10 * 3 bits = 30 bits |
| d | 10 | 010 | e: | 4 * 4 bits = 16 bits |
| e | 4 | 0110 | h: | 3 * 5 bits = 15 bits |
| h | 3 | 01110 | i: | 15 * 2 bits = 30 bits |
| i | 15 | 10 | j: | 4 * 4 bits = 16 bits |
| l | 4 | 1100 | r: | 1 * 6 bits = 6 bits |
| o | 1 | 111011 | r: | 3 * 5 bits = 15 bits |
| r | 3 | 01111 | s: | 4 * 4 bits = 16 bits |
| s | 4 | 1101 | t: | 2 * 5 bits = 10 bits |
| t | 2 | 11111 | space: | 11 * 3 bits = 33 bits |
| space | 11 | 001 | ∴ | 2 * 5 bits = 10 bits |
| . | 2 | 11100 | Total number of bits = 255 bits | |

- Because we set it to $4096 = 2^{12}$ entries, we can use 12 bits to represent the codes. Although in this case, 9 bits would have sufficed as 270 codes were used.